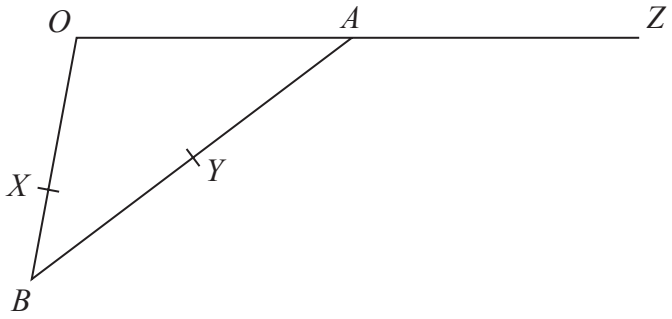
**21**




$OAB$ is a triangle.
$A$ is the midpoint of $OZ$
$Y$ is the midpoint of $AB$
$X$ is a point on $OB$

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OX} = 2\mathbf{b}$ $\overrightarrow{XB} = \mathbf{b}$

Prove that $XYZ$ is a straight line.

OB = 3b and OZ = 2a.

$\overrightarrow{BA}$ = a – 3b. Since Y is the midpoint of AB,

**1 mark**
Find BA or BY

$\overrightarrow{BY}$ = ½(a – 3b).

$\overrightarrow{XY}$ = XB + BY = b + ½(a – 3b)

= ½a – ½b = ½(a – b).

**1 mark**
Connected route to XY

$\overrightarrow{XZ}$ = XO + OZ = –2b + 2a = 2(a – b).

**1 mark**
Find a second vector

$\overrightarrow{XZ}$ = 4 $\overrightarrow{XY}$ .

**1 mark**
Scalar multiple

XY and XZ are parallel and share X, so X, Y and Z lie on a straight line.

**1 mark**
Parallel; common point therefore collinear

**(Total for Question 21 is 5 marks)**